

Module 3: Finding A Research Question - Key Terms & Definitions

Finding A Research Question

Comparative Research Question: aims to discover differences between groups within the context of an outcome variable.

- Clarifies the difference between ***one group with an outcome variable*** and ***another group without outcome variable***.
 - **Example:** *Is there a difference in the reduction of lung metastasis in osteosarcoma patients who received the vitamin D adjunctive therapy (group with outcome variable) compared with osteosarcoma patients who did not receive the vitamin D adjunctive therapy (group without outcome variable)?*
- ***Compares the effects of variables***

Example: *How does the vitamin D analogue 22-Oxacalcitriol (variable 1) mimic the antiproliferative activity of 1,25-Dihydroxyvitamin D (variable 2) in osteosarcoma cells?*

Constructivism: is a philosophical position used in **qualitative research**.

- The major elements are:
 - Understanding (i.e., the belief that individuals seek understanding of the world in which they live and work).
 - Multiple participant meanings (i.e., individuals develop subjective meanings of their experiences, which are varied and multiple).

- Social and historical construction (i.e., the meanings are formed through historical and social norms).
- Theory generation (i.e., rather than starting with a theory, as in **post-positivism**, inquirers generate or inductively develop a theory or pattern of meaning).

Descriptive Research Question: attempts to describe the behavior of a population in relation to one or more variables, or describe the characteristics of variables to be measured.

- Measures responses of subjects to variables
- Presents variables to measure, analyze, or assess
 - **Example:** *What is the proportion of resident doctors in the hospital who have mastered ultrasonography (response of subjects to a variable) as a diagnostic technique in their clinical training?*

Hypothesis: makes a specific prediction about a new phenomenon or a formal statement on the expected relationship between an independent variable and a dependent variable. A **hypothesis** provides a tentative answer to the **research question** to be tested or explored.

- **Well-constructed hypotheses** are *realistic, in-depth, based on original ideas, sufficiently complex, backed by preliminary evidence, reproducible, can be predicted, have evidence-based logical reasoning, and can be empirically tested.*

Table 1. Summary of types of quantitative and qualitative research questions and hypotheses

Quantitative research questions	Quantitative research hypotheses
Descriptive research questions	Simple hypothesis
Comparative research questions	Complex hypothesis
Relationship research questions	Directional hypothesis
	Non-directional hypothesis
	Associative hypothesis
	Causal hypothesis
	Null hypothesis
	Alternative hypothesis
	Working hypothesis
	Statistical hypothesis
	Logical hypothesis
	Hypothesis-testing
Qualitative research questions	Qualitative research hypotheses
Contextual research questions	Hypothesis-generating
Descriptive research questions	
Evaluation research questions	
Explanatory research questions	
Exploratory research questions	
Generative research questions	
Ideological research questions	
Ethnographic research questions	
Phenomenological research questions	
Grounded theory questions	
Qualitative case study questions	

Table 3. Definitions and examples of quantitative research hypotheses

Quantitative research hypotheses

Simple hypothesis

- Predicts relationship between single dependent variable and single independent variable

If the dose of the new medication (single independent variable) is high, blood pressure (single dependent variable) is lowered.

Complex hypothesis

- Foretells relationship between two or more independent and dependent variables

The higher the use of anticancer drugs, radiation therapy, and adjunctive agents (3 independent variables), the higher would be the survival (1 dependent variable).

Directional hypothesis

- Identifies study direction based on theory towards particular outcome to clarify relationship between variables

Privately funded research projects will have a larger international scope (study direction) than publicly funded research projects.

Non-directional hypothesis

- Nature of relationship between two variables or exact study direction is not identified

- Does not involve a theory

Women and men are different in terms of helpfulness. (Exact study direction is not identified)

Associative hypothesis

- Describes variable interdependency

- Change in one variable causes change in another variable

A larger number of people vaccinated against COVID-19 in the region (change in independent variable) will reduce the region's incidence of infection (change in dependent variable).

Causal hypothesis

- An effect on dependent variable is predicted from manipulation of independent variable

A change into a high-fiber diet (independent variable) will reduce the blood sugar level (dependent variable) of the patient.

Null hypothesis

- A negative statement indicating no relationship or difference between 2 variables

There is no significant difference in the severity of pulmonary metastases between the new drug (variable 1) and the current drug (variable 2).

Alternative hypothesis

- Following a null hypothesis, an alternative hypothesis predicts a relationship between 2 study variables

The new drug (variable 1) is better on average in reducing the level of pain from pulmonary metastasis than the current drug (variable 2).

Working hypothesis

- A hypothesis that is initially accepted for further research to produce a feasible theory

Dairy cows fed with concentrates of different formulations will produce different amounts of milk.

Statistical hypothesis

- Assumption about the value of population parameter or relationship among several population characteristics

- Validity tested by a statistical experiment or analysis

The mean recovery rate from COVID-19 infection (value of population parameter) is not significantly different between population 1 and population 2.

There is a positive correlation between the level of stress at the workplace and the number of suicides (population characteristics) among working professionals.

Logical hypothesis

- Offers or proposes an explanation with limited or no extensive evidence

If healthcare workers provide more educational programs about contraception methods, the number of adolescent pregnancies will be less.

Hypothesis-testing (Quantitative hypothesis-testing research)

- Quantitative research uses deductive reasoning.

This involves the formation of a hypothesis, collection of data in the investigation of the problem, analysis and use of the data from the investigation to draw conclusions to validate or nullify the hypotheses.

Mixed Methods Research: is an approach to inquiry *involving collecting both quantitative and qualitative data, using a specific procedure or design, combining (or*

integrating) the two forms of data with the design, and *drawing conclusions about the insight to emerge from the combined databases*. This description emphasizes a **methods perspective** focused on understanding mixed methods research from its data collection, data analysis, and interpretation. Also, in mixed methods a researcher brings philosophical assumptions and theories that inform the conduct of the research.

- Research Design, e.g., Convergent Mixed Methods Design, Explanatory Sequential Mixed Methods Design, Exploratory Sequential Mixed Methods Design, Complex Mixed Methods Design (for concrete definitions, see our reading, Creswell, p. 10).

Post-Positivism: See Social Sciences – Key Terms & Definitions.

- This view is sometimes called **scientific method** or doing **science research**.
- **Quantitative Research** is closely associated with the **Post-Positivism Worldview**.
- The major elements are:
 - Determination (i.e., causes determine effects or outcomes)
 - Reductionism (i.e., reduces the ideas into a small, discrete set to test, such as the variables that comprise hypotheses and research questions)
 - Empirical observation and measurement
 - Theory verification (i.e., theories need to be tested, verified, and refined).

Pragmatic Worldview: is concerned with applications – what works – and solutions to problems.

- The major elements are:
 - Consequence of actions
 - Problem-centered
 - Pluralistic (i.e., multiple methods, different worldviews, assumptions, and forms of data collection and analysis)

- Real-world practice-oriented

Qualitative Research: is an approach for exploring and understanding the meaning individuals or groups ascribe to a social or human problem. The research process involves emerging questions and procedures, data typically collected in the participant's setting, data analysis ***inductively*** building from particulars to general themes, and the researcher making ***interpretations of the meaning of the data***. The final written report has a flexible structure. Those who engage in this form of inquiry use an ***inductive style building from data to themes*** and ***a focus on individual meaning***, and ***emphasize the importance of reporting the complexity of a situation***.

- Research Designs: e.g., Ethnographies, Case Study, Grounded Theory, Phenomenology, Narrative Research, Descriptive Method (for concrete definitions, see our reading, Creswell, p. 9).

Quantitative Research: is an approach for testing objective theories by ***examining the relationships among variables*** or ***comparing among groups***. These variables, in turn, can be measured, typically on instruments, so that numbered data can be analyzed using ***statistical procedures***. The final written report has a set structure comprising an introduction, methods, results, and discussion. Theories are tested ***deductively***, built into a study to ***protect against bias, control for alternative or counterfactual explanations***, and seek to ***generalize*** and ***replicate the findings***.

- Research Designs, e.g., Experimental and Quasi-Experimental, Single-Subject, Nonexperimental, such as Surveys, Longitudinal (see more on it below).
- **Note:** The **Quantitative Approach** is the **best approach to test a theory or explanation**.

Relationship Research Question: elucidates trends and interactions among variables.

- Defines trends, associations, relationships, or interactions between dependent variable and independent variable.
 - **Example:** *Is there a relationship between the number of medical student suicides (dependent variable) and the level of medical student stress (independent variable) in Japan during the first wave of the COVID-19 pandemic.*

Research Approach: synonymously with **research methodology**. We looked into three broad methodologies: **Quantitative, Qualitative, and Mixed Methods**.

Table 1.5 Qualitative, Quantitative, and Mixed Methods Approaches

Tends to or Typically	Qualitative Approaches	Quantitative Approaches	Mixed Methods Approaches
Uses these philosophical assumptions	Constructivist/transformational knowledge claims	Postpositivist knowledge claims	Pragmatic knowledge claims
Employs these designs	Phenomenology, grounded theory, ethnography, case study, and narrative	Surveys and experiments (true, quasi-experimental, single-subject)	Sequential, convergent, and complex designs
Employs these methods	Open-ended questions, emerging approaches, text or image data	Closed-ended questions, predetermined approaches, numeric data (may include some open-ended questions)	Both open- and closed-ended questions, both emerging and predetermined approaches, and both quantitative and qualitative data and analysis
Uses these practices of research	Positions themselves Collects participant meanings Focuses on a single concept or phenomenon Brings personal values into the study Studies the context or setting of participants Validates the accuracy of findings Makes interpretations of the data Creates an agenda for change or reform Collaborates with the participants Employs text analysis procedures	Tests or verifies theories or explanations Identifies variables to study Relates variables in questions or hypotheses Uses standards of validity and reliability Observes and measures information numerically Uses unbiased approaches Employs statistical procedures	Collects both quantitative and qualitative data Develops a rationale for mixing Integrates the data at different stages of inquiry Presents visual pictures of the procedures in the study Employs the practices of both qualitative and quantitative research

Research Design: refers to the way of conducting research within a broad methodology.

We distinguished Research Design into three broad types: **Quantitative Design**,

Qualitative Design, and **Methods Design (Mixed Methods)**.

Quantitative Designs

- Experimental and quasi-experimental
- Single-subject
- Nonexperimental such as surveys
- Longitudinal

Qualitative Designs

- Descriptive method
 - Narrative research
 - Phenomenology
 - Grounded theory
 - Ethnography
 - Case study
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Research Methods: Within a particular **Research Design**, the researcher gathers data, analyzes it, and makes an interpretation of its meaning.

Quantitative Methods	Mixed Methods	Qualitative Methods
Predetermined	Both predetermined and emerging methods	Emerging methods
Instrument-based questions	Both open- and closed-ended questions	Open-ended questions
Performance data, attitude data, observational data, and census data	Multiple forms of data drawing on all possibilities	Interview data, observation data, document data, and audiovisual data
Statistical analysis	Statistical and text analysis	Text and image analysis
Statistical interpretation	Across databases interpretation	Themes, patterns, and interpretation

Research Question: is what a study aims to answer after data analysis and interpretation.

- A good research question is
 - feasible and interesting
 - manageable
 - novel
 - relevant
 - focused, specific, and concise
 - systematic
 - facilitates understanding of the research topic, objective scope, and the limitations of the study
 - requires a comprehensive literature search and an in-depth understanding of the problem being investigated
 - ethical
 - has a potential value/is publishable

Transformative Worldview: holds that research inquiry needs to be intertwined with

politics and a political change agenda to confront social oppression at whatever levels it occurs.

- The major elements are:
 - Political
 - Power and Justice oriented
 - Collaborative
 - Change oriented

Types of Quantitative Research Design:

- **Survey research:** provides a quantitative or numeric description of a population's trends, attitudes, or opinions by studying a sample of that population. It includes questionnaires or structured interviews for data collection to generalize from a sample to a population.
- **Experimental research:** seeks to determine if a specific treatment influences an outcome. The researcher assesses this by providing a specific treatment to one group, withholding it from another, and then determining how both groups scored on an outcome. It includes true experiments, the random assignment of subjects to treatment conditions, and quasi-experiments that use nonrandomized assignments.